

Amino Acid Transmitters

Glutamate; Neural Plasticity

1. Where in the brain do we find glutamatergic neurons?
2. Draw the enzymatic synthesis of glutamate.
3. Illustrate the inactivation of glutamatergic neurotransmission. How is this like or different from other inactivation mechanisms presented so far?
4. Name the 2 pharmacological classes of ionotropic glutamate (iGlu) receptors. For each:
 - give example(s) of an agonist(s) and antagonist(s)
 - name the ion(s) that permeate the pore of the receptor, and
 - describe their role in neurotransmission

Pharmacological class	AMPA Receptors	NMDA Receptors
Agonist		
Antagonist		
Permeable to which ions		
Role in neurotransmission		

5. What is LTP? How does it work?

Consider the following:

- depolarization
- NMDA receptors
- Ca^{++} influx
- role of kinases
- receptor trafficking